

ARKOPROBHO MITRA

Senior AI Engineer | GenAI, RAG & Agentic Systems | Technical Leadership

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PROFESSIONAL SUMMARY

AI Engineer and sub-team lead with nearly 3 years building GenAI and applied ML systems within 5+ years at Infosys. Led teams of two to four engineers across enterprise RAG, document intelligence, fraud detection, and multi-agent automation for regulated domains. Turns LLM prototypes into secure, evaluated, production-grade systems with measurable business impact.

TECHNICAL SKILLS

GenAI, RAG and agents: LLMs, LangGraph, LangChain, prompt engineering, tool calling, hybrid retrieval, RRF, Azure OpenAI (GPT-3.5 Turbo, GPT-4, GPT-4o), text-embedding-ada-002, Azure AI Search

Evaluation, security and LLM Ops: RAGAS, SME golden sets, Langfuse, CI/CD gates, Azure AD, Key Vault, human-in-the-loop, audit logging, MLflow, Evidently AI

ML and document intelligence: Python, scikit-learn, XGBoost, SHAP, LayoutLMv2, OCR, Azure Document Intelligence

Engineering and platforms: FastAPI, asyncio, PostgreSQL, Redis, Docker, GitHub Actions, Blob Storage, Elasticsearch, SAP Ariba, ServiceNow, SharePoint

PROFESSIONAL EXPERIENCE

Infosys | Aug 2021 - Present

AI Engineer (Sub-team Lead) | Topaz, Applied AI Practice | Kolkata, India | Nov 2023 - Present

Multi-Agent Procurement Copilot | Aug 2025 - Present

- Led technical delivery and mentored three AI engineers; reduced procurement cycle time by 35% and RFQ turnaround from three days to four hours. Transitioned during the Claims platform's planned closeout.
- Owned a five-agent LangGraph workflow (Planner, Supplier, Contract, RFQ, and Policy) using GPT-4o, with a separate escalation node for human review.
- Integrated SAP Ariba, ServiceNow, SharePoint, and contract RAG on Azure AI Search; used metadata access controls and blue-green index deployments.
- Evaluated 300 SME-authored cases at over 90% tool-selection accuracy and over 95% workflow completion; RAGAS and SME checks gated releases.

Insurance Claims Intelligence Platform | Nov 2024 - Sep 2025

- Directed four AI engineers and designed the document-understanding, fraud, explainability, and human-review architecture; reduced claim resolution from about four days to roughly two and a half days. Began during a planned Banking-to-Claims transition.
- Fine-tuned LayoutLMv2 on 15,000 labelled pages across 12 classes using Azure ML A100 compute; achieved 94% accuracy on a held-out 2,000-page test set.
- Trained XGBoost on 800,000 claims with 3% fraud prevalence. A 0.72 operating threshold raised precision from 41% to 78% while accepting lower recall; SHAP explained each flag.
- Used rules, GPT-3.5 Turbo, and GPT-4 tiering to cut LLM spend by 71% with under 3% quality impact; added Evidently AI monitoring and MLflow retraining.

Enterprise Knowledge Assistant - Global Banking | Nov 2023 - Dec 2024

- Led a two-engineer sub-team and owned retrieval, access controls, and evaluation for a GenAI assistant over a 2.5 million-document corpus; compliance-query time fell by about 60% in a pilot before-and-after study of 50 queries with 20 analysts, and the platform later scaled to 30,000+ monthly users.
- Delivered two phases: GPT-3.5 Turbo with BM25 and semantic search, then GPT-4 with hybrid vector plus BM25/RRF; precision improved 14% relative on regulation-code queries.
- Used text-embedding-ada-002 and cross-encoder reranking; a Redis semantic cache reduced p50 latency from 18 seconds to 3.8 seconds.
- Enforced pre-retrieval Azure AD access filters. A 200-query SME set, 89% RAGAS faithfulness, and CI/CD gates controlled releases.

Systems Engineer - Data and Logistics | Logistics and Data Systems Practice | India | Aug 2021 - Oct 2023

- Optimized backend data pipelines and Elasticsearch search infrastructure supporting daily logistics operations and real-time shipment search.
- Built QlikSense dashboards, maintained ServiceNow incident and change workflows, and supported client data migrations and legacy-system integrations.

RESEARCH PUBLICATION

How to handle bi/tri-lingual Indic texts in a single image? A new dataset of natural scene and born-digital images
Multimedia Tools and Applications, Springer, Vol. 81, pp. 15367-15394 (2022) | DOI: 10.1007/s11042-022-12596-7
Co-authored research that constructed and published a multilingual dataset of natural-scene and born-digital images containing Indic script combinations for OCR and scene-text detection in low-resource scripts.

SELECTED PORTFOLIO PROJECTS

Agentic Job Application Pipeline | 2026: LangGraph, FastAPI, and asyncio workflow for job discovery, LLM scoring, tailored application drafting, and human review.

LangGraph Fact-Checker with Human Review | 2026: LangGraph and FAISS verification pipeline in which an LLM verdict agent classifies claims as supported, refuted, or unverifiable with confidence scores; low-confidence cases route to human review, with Langfuse trace-level observability.

Full-stack E-Commerce Platform with RAG Assistant | 2025: React, Spring Boot, and FastAPI microservices platform with PostgreSQL, semantic search, review sentiment classification, RAG support, and distributed authentication.

EDUCATION AND CERTIFICATION

Bachelor of Engineering (B.E.) | Jadavpur University, Kolkata, India | 2017 - 2021

AWS Certified Cloud Practitioner | Amazon Web Services